

HIERARCHY MODELS

The Hierarchy Models define the layered structure of Huey's operating system, ensuring efficient task distribution and coordinated operation across strategic, operational, and real-time layers. This architecture is inspired by both natural and engineered systems.

MacroOS:

- Serves as the strategic brain, overseeing system-wide objectives and long-term planning.
- Issues high-level directives and allocates resources for maximum efficiency.

MicroOS:

- Functions as the operational backbone, translating strategic goals into actionable workloads and managing mid-level processes.
- Coordinates between MacroOS planning and NanoOS execution.

NanoOS:

- Operates as the precision worker, managing real-time interactions and low-level control.
- Responds to environmental inputs and immediate operational demands.

Advantages of the Hierarchical Approach:

- Scalability – Each layer can be expanded or upgraded independently.
- Resilience – Redundant processes at each layer safeguard against single points of failure.
- Flexibility – The system can adapt quickly to changing mission requirements.

Governance Context:

- Each layer interacts with governance protocols to ensure constitutional compliance.
- Decision-making authority is clearly defined and delegated based on operational scope.

Testing Objectives:

- Verify communication efficiency between layers under varying operational loads.
- Measure system adaptability in response to simulated disruptions at different layers.